Personalized Skill Gap Tracking & Recommendation System Using MongoDB

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🌟 Project Overview

The **Personalized Skill Gap Tracking and Recommendation System** is a full-stack, database-centric application designed to evaluate a learner's technical competencies against industry-standard benchmarks across **22 Computer Science career roles**.

Built natively on **MongoDB**, the system leverages advanced MongoDB features such as **Aggregation Pipelines (`$lookup`, `$unwind`, `$filter`, `$max`, `$multiply`)**, **Text Indexes**, and flexible document schemas to compute weighted skill gaps, calculate an overall **Career Readiness Score (0–100%)**, determine **Multi-Role Alternative Fit**, and provide **Curated Learning Roadmaps**.

🚀 Key Features

1. **Comprehensive CS Career Taxonomy**:
2. 22 Specialized roles spanning Software Engineering, AI & Machine Learning, Cloud & DevOps, Cybersecurity, Systems & IoT, Database Architecture, and Product Management.
3. Standardized 1–5 proficiency scale with importance weight factors.
4. **Server-Side MongoDB Aggregation Engine**:
5. Pure MongoDB pipeline computes skill gaps and priority urgency (`priorityScore = gap × weight`).
6. Dynamically joins recommended courses from the `recommendations` collection via `$lookup` .
7. **Career Readiness Score & Multi-Role Fit**:
8. Computes weighted overall qualification match %.
9. Cross-evaluates the candidate against all other CS roles to identify alternative career pathways.
10. **Smart Resume / Bio Skill Extractor**:

11. Paste resume summary or project descriptions; MongoDB text pattern matching extracts recognized skills to instantly populate candidate profiles.

12. **Interactive "What-If" Skill Simulator**:

13. Dynamic real-time sliders allow candidates to preview how leveling up specific skills impacts their overall readiness without altering database records.

14. **Historical Progress Logging (Objective 5)**:

15. Tracks skill upgrades over time with timestamped snapshots in `progressLogs`.

16. **One-Click Printable Skill Audit Report**:

17. Formats the full analysis into a clean, printable PDF / lab report sheet.

🛠️ Tech Stack & Architecture

- **Database**: MongoDB v7 (Running inside Docker container `skilltracker-mongo`)
- **Backend**: Node.js, Express.js (v5), TypeScript, MongoDB Node Driver (`mongodb`)
- **Frontend**: HTML5, Modern CSS (Tailwind CSS), JavaScript (ES6+), Chart.js
- **Containerization**: Docker Desktop (`mongo:7` on port `27017`)

💻 How to Run the Project (Step-by-Step Guide)

Anyone can clone and run this project in under **2 minutes**. Follow the steps below:

⚙️ Prerequisites

Ensure you have the following installed on your computer: 1. **Node.js (v18 or higher)**: Download Node.js 2. **MongoDB**: - **Option A (Local/Docker)**: Docker Desktop running `skilltracker-mongo` or local MongoDB Community Server on port `27017`. - **Option B (Cloud MongoDB Atlas)**: Zero-installation cloud database URI (see below).

Step 1: Clone or Download the Project

Using Git:

```
git clone https://github.com/THARAKESHA/personalized-skill-gap-tracker.git
cd personalized-skill-gap-tracker
```

OR Download as ZIP (No Git Required):

1. Click the green `<> Code` button at the top of this repository.
2. Click `Download ZIP`.
3. Extract the ZIP file and open the folder in your terminal or file explorer.

Step 2: Choose Your Running Method

🟢 Method A: The 1-Click Launch (Recommended for Windows)

If you are on Windows, simply **double-click the file:**

```
run.bat
```

> **What `run.bat` does automatically:** - Checks if Node.js is installed. - Automatically runs `npm install` if running for the first time. - Automatically runs `npm run seed` to populate roles and skills into MongoDB. - Automatically starts the MongoDB Docker container (if Docker is used). - Starts the web server and displays the live URL!

⌨️ Method B: Terminal / Command Line (Windows, macOS, Linux)

If you prefer using the terminal (or are on macOS/Linux), run these 3 standard commands:

1. Install Dependencies:

```
npm install
```

2. Seed the Database Collections (Roles, Skills & Curated Courses):

```
npm run seed
```

(You will see green checkmarks confirming 22 CS roles, 39 skills, and 25 courses are inserted).

3. Start the Web Server:

```
npm start
```

(Or use `npm run dev` for hot-reload development mode).

Expected Terminal Output:

```
Connected to MongoDB
=====================================================
Skill Gap Recognition Server running on http://localhost:4000
Interactive Web UI accessible at http://localhost:4000
=====================================================
```

Step 3: Open the Web Dashboard

Open your web browser (Chrome, Edge, Firefox) and navigate to:

👉 `http://localhost:4000`

🌐 Running with Free Cloud MongoDB Atlas (No Docker Needed)

If you (or your evaluator) do not have Docker or local MongoDB installed: 1. Create a free cluster on MongoDB Atlas. 2. Create a `.env` file in the project root (copied from `.env.example`). 3. Add your Atlas connection string: `env MONGODB_URI=mongodb+srv://<username>:<password>@cluster0.mongodb.net/skilltracker?retryWrites=true&w=majority` 4. Run `npm run seed` once to populate the cloud database, then run `npm start`.

🎯 Quick Interactive Walkthrough (What to Test):

1. **Select an Active Profile**: In the top navigation bar, choose from sample personas (**Alex Kumar**, **Priya Sharma**, **Rohan Verma**, or **Tharakesha**), or click `+ New User` to register your own profile.
2. **Select Target Career Role**: Change the career dropdown to any of the 22 CS roles (e.g. *Full Stack Developer*, *AI/ML Engineer*, *DevOps Engineer*, *Cybersecurity Analyst*) to watch the MongoDB aggregation engine recalculate gaps in real-time.
3. **Inspect the Visual Benchmark**: Review the Chart.js side-by-side bar chart (Green = Current Level vs. Purple = Benchmark Required).
4. **Explore Alternative Career Match**: Look at the left panel to see alternative career pathways your current skillset already qualifies you for.
5. **Resume / Bio Extractor**: Click `Resume Extractor` in the header, paste a short project or resume description, and click `Scan & Extract` to see MongoDB's text indexing extract verified skills automatically!
6. **"What-If" Skill Simulator**: Click `What-If Simulator` in the top bar to drag sliders and preview how learning specific skills impacts your overall readiness score.
7. **Print Audit Report**: Click `Print Report` to generate a clean, formatted audit sheet ready for lab submission.
8. **To Stop the Server**: Press `Ctrl + C` in the command prompt window.

📡 REST API Reference

Method	Endpoint	Description
`GET`	`/api/health`	System and MongoDB connection status
`GET`	`/api/users`	List all candidate profiles
`POST`	`/api/users`	Create a new candidate profile
`PUT`	`/api/users/:id`	Update candidate skills or target career
`GET`	`/api/roles`	List all 22 CS career benchmark roles
`GET`	`/api/skills`	List master standardized skill taxonomy
`POST`	`/api/skills/extract`	Auto-extract skills from resume/bio text
`GET`	`/api/skill-gap/:userId`	Full aggregation gap audit, readiness %, and course recommendations
`POST`	`/api/skill-gap/simulate`	In-memory "What-If" simulator recalculation
`POST`	`/api/progress/:userId/level-up`	Upgrade a skill level and log historical progress snapshot
`GET`	`/api/progress/:userId`	Get historical progression logs

📂 Project Directory Structure

```
skill-gap-tracker/
├── public/
│   └── index.html               # Modern Single-Page Dashboard & Report UI
├── src/
│   ├── data/
│   │   ├── roles.json           # 22 Computer Science career benchmarks
│   │   ├── skills.json          # 39 Standardized technical competencies
│   │   ├── recommendations.json # Curated learning resources & courses
│   │   └── users.json           # Sample testing personas
│   ├── routes/
│   │   ├── skillGap.ts          # Core MongoDB aggregation pipeline & simulation
│   │   ├── skills.ts            # Master skills & smart resume extractor
│   │   ├── progress.ts          # Progress snapshots & level-up tracking
│   │   ├── users.ts             # Candidate profile management
│   │   └── targetRoles.ts       # Career roles repository
│   ├── db.ts                    # MongoDB connection pool setup
│   ├── index.ts                 # Express server configuration
│   └── seed.ts                  # Master automated database seeder
├── package.json
├── tsconfig.json
└── README.md
```